# Pipes in the Desert
# GUI Program
E5 - TEAM A

Supervisor:
Dr. Balla Katalin


**Members:**

| | | |
|---|---|---|
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# 13. Complete program

## 13.1 Deployment guide

### 13.1.1 List of files

| File name | Size | Date | Content |
|---|---|---|---|
| Cistern.java | 6.6 KB | 24.05.2026 | Handles cistern behavior and stored materials. |
| FinalGuiMain.java | 120.3 KB | 25.05.2026 | Starts and manages the final GUI program. |
| GameTimer.java | 2.6 KB | 24.05.2026 | Handles the game countdown timer. |
| NetworkElement.java | 4.2 KB | 24.05.2026 | Base class of pipe-system elements. |
| Pipe.java | 7.9 KB | 25.05.2026 | Handles pipe state, water, puncture, repair, and occupancy. |
| PipeNetwork.java | 17.4 KB | 25.05.2026 | Stores and updates the pipe network. |
| Player.java | 5.3 KB | 24.05.2026 | Base class for players. |
| Plumber.java | 7.5 KB | 25.05.2026 | Handles plumber actions. |
| Pump.java | 11.0 KB | 24.05.2026 | Handles pump state, direction, repair, and breakdown. |
| Saboteur.java | 1.9 KB | 24.05.2026 | Handles saboteur actions. |
| ScoreBoard.java | 3.2 KB | 24.05.2026 | Stores plumber and saboteur scores. |
| Spring.java | 3.2 KB | 24.05.2026 | Produces water for the network. |
| System.java | 11.9 KB | 24.05.2026 | Controls game setup, turns, and game state. |
| Team.java | 2.2 KB | 24.05.2026 | Stores team members. |
| WaterFlowManager.java | 7.1 KB | 25.05.2026 | Calculates water flow and score changes. |
| Main.java | 28.7 KB | 24.05.2026 | Console entry point and prototype menu version of the program. |

### 13.1.2 Compilation

The program was compiled using Java JDK 11 and Maven. The compilation must be started from the root folder of the project, where the **pom.xml** file is located.

Command:

**mvn clean compile**

This command compiles all Java source files and places the generated **.class** files into the **target/classes** folder.

If Maven is not used, the program can also be compiled directly with:

**javac -encoding UTF-8 -d target\classes src\main\java\app*.java**

### 13.1.3 Run

The final graphical version of the program is started from the project root folder. No additional command-line parameters are required.

Command using Maven:

**mvn exec:java**

Alternative command after compilation:

**java -cp target\classes app.FinalGuiMain**

After starting the program, the main menu of **Pipes in the Desert** appears. From there, the user can start a new game, read the instructions, or exit the program.

## 13.2 Evaluation

| Name of the team member | Participation (%) |
|---|---|
| Muhammad Ibrahim Shoeb | 16.6% |
| Muhammad Hameez Khan | 16.6% |
| Arda Gecegörür | 16.6% |
| Ilgın Tunç | 16.6% |
| Yahya Akhrikhar | 16.6% |
| Aasif Mohd | 16.6% |

## 13.3 Tools Used

| Tool | Purpose | Description |
|---|---|---|
| ChatGPT (AI) | Writing assistance | Used for proofreading, grammar correction, brainstorming, and clarifying design decisions. Also used for AI traceability check. |

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| VS Code | Development environment | Used for writing, editing, debugging, compiling, and organizing the Java source code during implementation of the project. |
| Microsoft Word | Documentation | Used to write, format, and organize the GUI specification document. |
| Discord | Communication | Used for online team meetings and coordination. |
| Google Meet | Communication | Used for real-time collaboration during working sessions. |
| GitHub | Version control | Used to manage and share source code and documentation. |
| Microsoft Teams | Communication | Used for file sharing and informal coordination. |

## 13.3 Protocol

| Start (date & time) | Duration | Performer(s) | Activity description |
|---|---|---|---|
| 20.05.2026 17:00 | 3 hours | All members | Meeting regarding the implementation of the graphical user interface for the complete program. The team reviewed the previously prepared GUI specification document and discussed how the planned interface components would be integrated into the existing prototype system. Decisions were made about the implementation responsibilities of GUI windows, rendering logic, player interaction handling, and synchronization between GUI events and the game model. The team also agreed on maintaining consistency between the final implementation and the previously documented MVC-based GUI architecture. |

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| 21.05.2026 20:30 | 9 hours | Muhammad Ibrahim Shoeb, Yahya Akhrikhar, Ilgın Tunç, Arda Gecegörür | Meeting and collaborative implementation session for GUI integration. GUI elements including menus, game board visualization, player interaction controls, and game status displays were implemented and connected to the existing game logic classes. Rendering updates, button interactions, and visual feedback mechanisms were integrated into the program. The team also performed initial integration testing during implementation to verify that GUI actions correctly triggered prototype game methods and updated the visual state consistently. |
| 22.05.2026 14:00 | 10 hours | Muhammad Ibrahim Shoeb | Individual implementation work for the complete GUI program. The activity focused on building and improving the final graphical interface, adding code logic for GUI interactions, defining animation boundaries, and ensuring that the final program remained consistent with the documented requirements and design decisions prepared throughout the semester. |
| 23.05.2026 12:00 | 6 hours | Muhammad Hameez Khan | Individual testing activity for the complete program. Multiple testing sessions were performed, including smoke tests, analytical tests, and edge-case testing. The activity focused on discovering hidden or hard-to-detect bugs, checking unusual gameplay situations, and refining behavior in cases that were ambiguous or not explicitly defined in the original problem statement. |
| 23.05.2026 21:00 | 3 hours | Muhammad Hameez Khan | Individual testing and bug-fixing activity for the complete program. The interaction between GUI components and model classes was tested under different gameplay situations. Bugs related to game flow, event handling, and visual synchronization were corrected. Additional validation was performed for edge cases and invalid player actions to ensure stable runtime behavior. |

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| 24.05.2026 14:00 | 3 hour | Muhammad Hameez Khan | Individual documentation and final evaluation activity. The final program was reviewed from the documentation perspective, including the evaluation of implemented functionality, consistency with previous project documents, and preparation of the final wrap-up text for the Complete Program documentation. |
| 24.05.2026 18:00 | 2 hours | Muhammad Hameez Khan | Formatting and final organization of the Complete Program document. Section structure, formatting consistency, table layouts, captions, and terminology were reviewed and corrected. The deployment guide, evaluation section, and protocol entries were checked for consistency with the implemented complete program and previous project documentation. |
| 24.06.2026 20:00 | 2 hours | Arda Gecegörür | Performed the AI-based reverse traceability check between the implemented final program and previous project documents. The implemented classes, GUI actions, and system features were traced back to the problem definition, requirements specification, analysis model, prototype plans, and GUI documentation. Partial inconsistencies and implementation-specific additions were identified and documented for the final evaluation section. |
| 25.05.2026 19:00 | 1.5 hours | All members | Final review meeting before submission. The team reviewed the completed implementation, verified that the GUI and game logic behaved consistently with the documented specifications, and confirmed that all required files and documentation sections were complete. Final corrections were applied to the codebase and documentation before submission. |